As a pathologist, I studied disease through a microscope. As an epidemiologist, I studied patterns across communities.

Then it became personal.

After I lost my job, my family lost our health insurance. The next month, my husband was bitten by a tick and became ill with Lyme disease.

For immigrants like my husband, language can make tests, bills, and medical visits harder to understand.

I'm Yuwei Zhang. That experience led me to build PathwayAI.

The burden starts before a diagnosis. Finding care, missing work, arranging transport, and relying on family all take a toll. PathwayAI connects practical patient tools with evidence for community action.

Here is a fictional example. Someone paid two hundred twenty-five dollars for care and travel, missed three workdays, and needed eight hours of unpaid help.

One record shows money, missed work, and unpaid help, ready for a care or support conversation.

We compare Dutchess and Orange using CDC, Census, HRSA, CDC PLACES, CMS, New York surveillance, and published research. Together, they connect reported disease, community needs, healthcare resources, and spending context.

Each source answers a different question. Lyme cost illustrations and Medicare payments are clearly separated from personal bills and measured county Lyme spending.

Now let me show you the pathway.

Choose Quick and easy. Short prompts help on difficult days. Switch to More detail without losing your notes.

Before a bite, prepare for time outdoors. After a bite, find removal steps and time-sensitive guidance to discuss with a clinician.

Without insurance, where do you start? PathwayAI links to community health centers and helps people ask about income-based fees, coverage, and hospital financial assistance. Care is not automatically free; the tool gives people a practical place to begin.

Next, prepare for a visit. Our fictional patient describes a tick encounter, symptoms two weeks later, and a blood test they report was positive.

The editable timeline keeps important clues easy to find. Symptom onset and diagnosis delay remain separate. Review, correct, download, and bring the original test report.

People can write in their preferred language. With permission, optional AI organizes and translates an English summary beside the original for review. It supports preparation and does not replace a qualified medical interpreter.

English notes also work without external AI. AI organizes language. Arithmetic calculates costs. The patient remains at the center.

Patients can also record an unpaid balance and prepare questions about bill assistance or transport eligibility, cost, and booking.

For county teams, the planning brief connects a barrier with a possible response, partners, resources, and follow-up.

For example: could a small care-navigation pilot help uninsured residents reach existing services? Partners can discuss staffing, verified contacts, and referral follow-up. A pilot could track appointments reached, assistance received, and unresolved needs.

Partners get a concrete proposal to assess against local capacity and funding.

Patients and caregivers can use these tools today. We are seeking county partners to test the planning value.

People living with Lyme disease and a county health commissioner reviewed the working app. Their feedback led to earlier reading choices, shorter instructions, and clearer next steps. Sixty-eight automated tests passed. Patient reviews guide usability; software tests check behavior. Clinical accuracy, improved access, and savings still need evaluation.

The pilot is available on Streamlit. Our next step is a small pilot recruited through willing patient groups, health centers, and care navigators.

We propose free patient access supported by grants and community partnerships. Funding is not yet secured.

We will test usefulness, verify resources, and follow support connections. Shared patient-report collection remains off until consent and secure storage are ready. Longer-term records could help reveal burden that single visits miss.

We welcome partners to help turn patient experience into practical support.

PathwayAI: Full Economic Burden. Making the hidden patient journey visible, and actionable.

Thank you to the patients and experts helping make it possible.